Data Analytics using Python

Sessional Answers

1.a) List and explain Measures of Central Tendency

Ans.1.a) Measures of central tendency are statistical measures that provide information about the central location or typical value of a dataset. There are three main measures of central tendency:

1. **Mean**: The mean is the average of a dataset and is calculated by summing all the values in the dataset and dividing the sum by the number of values in the dataset. It is applicable for interval and ratio data and is affected by each value in the dataset, including extreme values.

   Example: For the dataset {2, 5, 7, 9, 11}, the mean is (2 + 5 + 7 + 9 + 11) / 5 = 7.4.

2. **Median**: The median is the middle value in an ordered array of numbers and is applicable for ordinal, interval, and ratio data. It is unaffected by extremely large and extremely small values. The median's position in an ordered array is given by (n+1)/2, where n is the number of terms in the dataset.

   Example: For the dataset {2, 5, 7, 9, 11}, the median is 7.

3. **Mode**: The mode is the most frequently occurring value in a dataset and is applicable to all levels of data measurement, including nominal, ordinal, interval, and ratio data. A dataset can have more than one mode, in which case it is called bimodal or multimodal.

   Example: For the dataset {2, 5, 7, 7, 9, 11}, the mode is 7.

In summary, the mean is the average of all the values in the dataset, the median is the middle value in an ordered array of numbers, and the mode is the most frequently occurring value in a dataset.

These measures of central tendency are used to summarize and describe the characteristics of a dataset and are used in various applications, including data analysis, statistics, and machine learning.

1.b) Define data analytics and its types

Ans.1.b) Data analytics is the scientific process of transforming data into insights for making better decisions. It involves the use of data, information technology, statistical analysis, quantitative methods, and mathematical or computer-based models to help managers gain improved insight about their business operations and make better, fact-based decisions.

There are four major types of data analytics:

1. **Descriptive analytics:** This is the conventional form of Business Intelligence and data analysis. It seeks to provide a depiction or "summary view" of facts and figures in an understandable format. It can summarize raw data and convert it into a form that can be easily understood by humans. Descriptive analysis or statistics can describe in detail about an event that has occurred in the past.

2. **Diagnostic analytics:** This is a form of advanced analytics which examines data or content to answer the question "Why did it happen?" Diagnostic analytical tools aid an analyst to dig deeper into an issue so that they can arrive at the source of a problem.

3. **Predictive analytics:** This helps to forecast trends based on the current events. Predicting the probability of an event happening in future or estimating the accurate time it will happen can all be determined with the help of predictive analytical models.

4. **Prescriptive analytics:** This indicates the best course of action to optimize the outcome. The goal of prescriptive analytics is to enable quality improvements, service enhancements, cost reductions, and increasing productivity.

Data analytics is important because it can help organizations to determine credit risk, develop new medicines, find more efficient ways to deliver products and services, prevent fraud, uncover cyber threats, retain the most valuable customers, and make better decisions based on data-driven insights.

2.a) List and explain various non-random Sampling Techniques.

Ans.2.a) Non-random sampling techniques are methods used to select a sample from a population without providing each member of the population with an equal chance of being selected. These methods are used when random sampling is not feasible or practical.

Here are some common non-random sampling techniques:

1. **Convenience Sampling**: Sample elements are selected based on their availability and accessibility to the researcher. This method is often used when time or resources are limited.

2. **Judgment Sampling**: Sample elements are selected based on the researcher's judgment and expertise. This method is often used when the researcher has prior knowledge or experience with the population.

3. **Quota Sampling**: Sample elements are selected based on specific quotas or criteria, such as age, gender, or income. This method is often used in market research and opinion polling.

4. **Snowball Sampling**: Sample elements are selected based on referrals from other sample elements. This method is often used when the population is difficult to reach or when the researcher is studying a hidden or hard-to-reach population.

These non-random sampling techniques are widely used in research and statistical analysis, but they are not as reliable as random sampling techniques because they are more susceptible to bias and may not provide a representative sample of the population.

2.b) Define the following: Probability - Terminology

i) **Experiment**
ii) **Event**
iii) **Elementary Events**
iv) **Sample Space**
v) **Unions and Intersections**
vi) **Mutually Exclusive Events**
vii) **Independent Events**
viii) **Collectively Exhaustive Events**
ix) **Complementary Events**

Ans.2.b) Probability - Terminology Definitions

i) Experiment:

- An experiment is a process that produces outcomes where there is more than one possible outcome, and only one outcome occurs per trial.

ii) Event:

- An event is an outcome of an experiment, which can be either an elementary event (indivisible) or an aggregate of elementary events.

iii) Elementary Events:

- Elementary events are outcomes that cannot be further decomposed or broken down into other events.

iv) Sample Space:

- The sample space is the set of all elementary events for an experiment, providing a comprehensive description of all possible outcomes.

v) Unions and Intersections:

- **Unions**: The union of two sets contains an instance of each element from both sets.
- **Intersections**: The intersection of two sets contains only those elements common to both sets.

vi) Mutually Exclusive Events:

- Mutually exclusive events are events with no common outcomes, where the occurrence of one event precludes the occurrence of the other event.

vii) Independent Events:

- Independent events are events where the occurrence of one event does not affect the occurrence or non-occurrence of the other event.

viii) Collectively Exhaustive Events:

- Collectively exhaustive events contain all elementary events for an experiment, ensuring that every possible outcome is covered.

ix) Complementary Events:

- Complementary events consist of all elementary events not in a specific event 'A,' representing the outcomes that are not part of event 'A.'

These definitions provide a comprehensive understanding of the key terminology related to probability, helping to clarify the fundamental concepts involved in probability theory.

3.a) What is regression analysis?

Ans.3.a) Regression analysis is a statistical method used to examine the relationship between a dependent variable and one or more independent variables. It aims to understand how the value of the dependent variable changes when one or more independent variables are varied. By fitting a regression model to the data, it helps in predicting the value of the dependent variable based on the independent variables.

The regression model is represented as y = b0 + b1x + e, where y is the dependent variable, x is the independent variable, b0 is the y-intercept, b1 is the slope, and e is the error term. The equation that describes how y is related to x and an error term is called the regression model. The equation that estimates the expected value of y for a given x value is called the simple linear regression equation.

The least squares method is used to estimate the regression coefficients, b0 and b1, by minimizing the sum of the squared errors between the observed and estimated values of y. The estimated regression equation is represented as ^ y = b0 + b1x, where ^ y is the estimated value of y for a given x value.

Regression analysis is used in various fields, including engineering, science, and business, to explore relationships between variables, optimize processes, and control processes. It is a powerful tool for making predictions and understanding the impact of different factors on an outcome.

3.a) Give Simple Linear Regression Model.

Ans.3.a) The Simple Linear Regression Model is a statistical method used to model the relationship between a dependent variable (y) and an independent variable (x). The model is represented as y = b0 + b1x + e, where y is the dependent variable, x is the independent variable, b0 is the y-intercept, b1 is the slope, and e is the error term. The equation that describes how y is related to x and an error term is called the regression model. The equation that estimates the expected value of y for a given x value is called the simple linear regression equation.

The slope (b1) represents the change in y for a one-unit increase in x. The y-intercept (b0) represents the expected value of y when x = 0. The error term (e) represents the random variation in y that is not explained by the model.

The Simple Linear Regression Model is used to understand the relationship between two variables, predict the value of the dependent variable based on the independent variable, and test hypotheses about the relationship between the variables. It is widely used in various fields, including economics, finance, and social sciences, to make predictions and understand the impact of different factors on an outcome.

In summary, the Simple Linear Regression Model is a statistical method used to model the relationship between a dependent variable and an independent variable. It estimates the expected value of the dependent variable based on the independent variable, and it is widely used in various fields to make predictions and understand the impact of different factors on an outcome.

3.b) What do you mean by one-way ANOVA?

Ans.3.b) One-way ANOVA, or one-way analysis of variance, is a statistical method used to compare the means of three or more groups to determine if there are statistically significant differences between them. In one-way ANOVA, there is one independent variable (factor) that divides the dataset into different groups, and the analysis focuses on whether there are differences in the means of the dependent variable across these groups.

The key concept in one-way ANOVA is to test whether the variation between the group means is significantly greater than the variation within the groups. This is done by comparing the variance between the group means to the variance within the groups. If the variation between the group means is significantly larger than the variation within the groups, it suggests that there are significant differences in the means of the groups.

In the context of the provided information, one-way ANOVA is used to test for the equality of three or more population means. It is applied when data from observational or experimental studies are available, and the goal is to determine if all population means are equal or if at least two population means have different values. The analysis involves setting up hypotheses, such as $H0: \mu1 = \mu2 = \mu3 = ... = \mu k$ (all population means are equal) and Ha: Not all population means are equal.

Overall, one-way ANOVA is a powerful statistical tool for comparing means across multiple groups and determining if there are significant differences in the population means.

3.c) Explain the principle behind the least square method.

Ans.3.c) The least squares method is a statistical technique used to estimate the parameters of a linear regression model. It is based on the principle of minimizing the sum of the squared differences between the observed values of the dependent variable and the predicted values based on the independent variable.

In the context of the air traffic controller stress test, the least squares method is used to estimate the parameters of the Simple Linear Regression Model, which describes the relationship between the stress level of a controller (dependent variable) and the workstation alternative they are assigned to (independent variable). The method involves finding the values of the slope (b1) and the y-intercept (b0) that minimize the sum of squared errors (SSE) between the observed stress levels and the predicted stress levels based on the workstation alternatives.

The formula for the least squares method involves calculating the sum of squares and sum of cross-products of the independent and dependent variables, as well as the mean values of the independent and dependent variables. The slope (b1) and y-intercept (b0) are then calculated using these values.

Once the estimated regression equation is obtained, it can be used to predict the stress level of a controller based on the workstation alternative they are assigned to. The estimated regression equation is also used to test hypotheses about the relationship between the variables, such as whether there is a significant difference in the mean stress levels among the three work station alternatives.

4.a) Give the three Approaches for Hypothesis Testing.

Ans.4.a) The three approaches for hypothesis testing are:

1. **P-Value Approach**:
    - The p-value is the probability, computed using the test statistic, that measures the support or lack of support provided by the sample for the null hypothesis.
    - If the p-value is less than or equal to the level of significance α, the null hypothesis is rejected.
    - The rejection rule is to reject H0 if the p-value is less than α.

2. **Critical Value Approach**:
    - The critical value approach involves using critical values from a standard normal distribution or t-distribution to determine the rejection region.
    - The test statistic is compared to the critical value to decide whether to reject the null hypothesis.
    - The rejection rule is based on comparing the test statistic to the critical value.

3. **Confidence Interval Approach**:
    - In this approach, a confidence interval is constructed for the population parameter of interest.
    - If the hypothesized value falls within the confidence interval, the null hypothesis is not rejected.
    - If the hypothesized value is outside the confidence interval, the null hypothesis is rejected.

These approaches provide different ways to assess the evidence from sample data and make decisions regarding the null hypothesis in hypothesis testing.

4.b) What is Hypothesis Testing? Define Null hypothesis and Alternate hypothesis.

Ans.4.b) Hypothesis Testing is a statistical method used to determine whether a statement about the value of a population parameter should or should not be rejected based on sample data. It involves formulating two competing statements, the null hypothesis (H0) and the alternative hypothesis (Ha) and using data from a sample to test these hypotheses.

The null hypothesis, denoted by H0, is a tentative assumption about a population parameter. It is typically a statement of no effect or no difference and is often the hypothesis that a researcher aims to disprove.

The alternative hypothesis, denoted by Ha, is the opposite of what is stated in the null hypothesis. It is the statement that a researcher hopes to support with evidence from the sample data.

In some cases, it is easier to identify the alternative hypothesis first, while in other cases the null hypothesis is easier. The context of the situation is very important in determining how the hypotheses should be stated.

For example, in a study to test whether a new manufacturing method is better than the current method, the alternative hypothesis might be "The new manufacturing method is better," while the null hypothesis would be "The new method is no better than the old method."

In general, a hypothesis test about the value of a population mean μ must take one of the following three forms:

One-tailed (lower-tail): H0: $\mu \geq \mu0$, Ha: $\mu < \mu0$

One-tailed (upper-tail): H0: $\mu \leq \mu0$, Ha: $\mu > \mu0$

Two-tailed: H0: $\mu = \mu0$, Ha: $\mu \neq \mu0$

where $\mu0$ is the hypothesized value of the population mean.

Hypothesis testing is used to determine whether the sample data provide sufficient evidence to reject the null hypothesis in favor of the alternative hypothesis. This is done by calculating a test statistic and comparing it to a critical value or a p-value. If the test statistic falls in the rejection region, the null hypothesis is rejected in favor of the alternative hypothesis. If the test statistic does not fall in the rejection region, the null hypothesis is not rejected.

It is important to note that hypothesis testing does not prove the alternative hypothesis to be true, but rather provides evidence against the null hypothesis.

4.c) Differentiate between type I and type II error.

Ans.4.c) Type I and Type II errors are two possible errors that can occur in hypothesis testing.

Type I error is rejecting the null hypothesis when it is actually true. This is also known as a false positive and is denoted by α. The probability of making a Type I error is called the level of significance of the test.

Type II error is accepting the null hypothesis when it is actually false. This is also known as a false negative and is denoted by β. The probability of making a Type II error is called the beta error.

The following table summarizes the differences between Type I and Type II errors:

	Null Hypothesis (H0) is true	Null Hypothesis (H0) is false
Correct Decision	Accept H0	Reject H0
Type I Error	Reject H0	Accept H0
Type II Error	Accept H0	Reject H0

	Type I Error (α)	Type II Error (β)
Definition	Rejecting H0 when it is true	Accepting H0 when it is false
Probability	Level of significance (α)	Beta error (β)
Consequence	False positive	False negative
Example	Convicting an innocent person	Failing to detect a defective product

In summary, Type I error is rejecting a true null hypothesis, while Type II error is accepting a false null hypothesis. The level of significance (α) is the probability of making a Type I error, while the beta error (β) is the probability of making a Type II error. The consequences of these errors can be significant, and it is important to carefully consider the potential risks and costs associated with each type of error.

5.a) **Define the terms:**

i. **Confidence interval**

ii. **Prediction interval**

iii. **Point Estimate**

Ans.5.a) i. **Confidence Interval**

A **confidence interval** is an interval estimate of the mean value of a dependent variable y for a given value of an independent variable x. It provides a range within which the true mean value of y is expected to lie with a certain level of confidence. In statistical terms, it represents the precision of estimating the average value of y for a specific x value. The margin of error for a confidence interval is smaller compared to a prediction interval, as it focuses on estimating the mean value of y rather than individual values.

ii. **Prediction Interval**

A **prediction interval** is an interval estimate used to predict an individual value of a dependent variable y for a given value of an independent variable x. Unlike a confidence interval that estimates the mean value of y, a prediction interval provides a range within which an individual value of y is expected to fall with a certain level of confidence. Prediction intervals are wider than confidence intervals due to the additional variability associated with predicting individual values rather than the mean value.

iii. **Point Estimate**

A **point estimate** is a single value that serves as an estimate of a parameter, such as the mean value of a dependent variable for a specific value of an independent variable. It is derived from a regression equation and represents the best guess for the average value of the dependent variable for a given independent variable value. In the context of regression analysis, a point estimate is used to predict either the mean value of y for a particular x or an individual value of y corresponding to a specific x value.

5.b) What is residual analysis ? Explain different types of residual analysis.

Ans.5.b) Residual Analysis

Residual analysis is a graphical examination of the residuals, which are the differences between the observed values of the dependent variable and the predicted values based on the regression model. The purpose of residual analysis is to validate the assumptions of the regression model and ensure that the model is appropriate for the data. The following are different types of residual analysis:

1. **Residual Plot Against x**: This plot shows the residuals against the values of the independent variable x. The assumption is that the variance is the same for all values of x. If the variability about the regression line is greater for larger values of x, the assumption of a constant variance of e is violated.

2. **Residual Plot Against Predicted Values**: This plot shows the residuals against the predicted values of the dependent variable. The pattern of this residual plot should be the same as the pattern of the residual plot against the independent variable x.

3. **Standardized Residuals**: Standardized residuals are obtained by dividing each residual by its standard deviation. A random variable is standardized by subtracting its mean and dividing the result by its standard deviation.

4. **Studentized Residual**: The standardized residual plot can provide insight about the assumption that the error term e has a normal distribution. If this assumption is satisfied, the distribution of the standardized residuals should appear to come from a standard normal probability distribution.

5. **Normal Probability Plot**: Another approach for determining the validity of the assumption that the error term has a normal distribution is the normal probability plot. To show how a normal probability plot is developed, we introduce the concept of normal scores.

Residual analysis is based on an examination of graphical plots, and it is the primary tool for determining whether the assumed regression model is appropriate. The assumptions provide the theoretical basis for the t test and the F test used to determine whether the relationship between x and y is significant, and for the confidence and prediction interval estimates. If the assumptions about the error term appear questionable, the hypothesis tests about the significance of the regression relationship and the interval estimation results may not be valid.

6.a) Explain the Normality of Residual (Error) in both linear and logistic regression.

Ans.6.a) In both linear regression and logistic regression, the normality of residuals (errors) is an important assumption that affects the validity of the models. Here is an explanation of the normality of residuals in both types of regression:

Linear Regression:

- **Normality of Residuals:** In linear regression, the normality of residuals assumption states that the residuals (the differences between the observed values and the values predicted by the model) should follow a normal distribution.

- **Importance:** Normality of residuals is crucial because it ensures that the statistical inferences made from the model, such as confidence intervals and hypothesis tests, are valid.

- **Checking Normality:** Normality of residuals can be checked visually using a Q-Q plot (Quantile-Quantile plot) or statistically using tests like the Shapiro-Wilk test.

- **Consequences of Violation:** If the normality assumption is violated, it can lead to biased parameter estimates, incorrect standard errors, and inaccurate hypothesis testing results.

Logistic Regression:

- **Normality of Residuals:** In logistic regression, the assumption of normality of residuals does not apply because logistic regression models the relationship between the predictors and the log-odds of the dependent variable, not the dependent variable itself.

- **Residuals in Logistic Regression:** In logistic regression, residuals are not normally distributed because the model does not assume a normal distribution for the dependent variable.

- **Checking Residuals:** Instead of normality, logistic regression focuses on other assumptions like linearity of log-odds and independence of errors.

- **Consequences of Violation:** While normality of residuals is not a concern in logistic regression, violations of other assumptions like linearity or independence can still impact the validity of the model.

In summary, while normality of residuals is a critical assumption in linear regression to ensure the validity of statistical inferences, it is not applicable in logistic regression due to the nature of the model and the way it handles the relationship between predictors and the dependent variable. In logistic regression, other assumptions take precedence, making the normality of residuals less relevant.

6.b) What do you mean by logistic regression and explain when it is used?

Ans.6.b) Logistic regression is a statistical modeling technique used to predict the probability of a binary or categorical outcome variable based on one or more predictor variables. It is particularly useful when the dependent variable is dichotomous, meaning it can only take two possible values, such as "yes/no", "success/failure", or "approved/rejected".

The logistic regression model has the form:

p = 1 / (1 + e^(-z))

The key aspects of logistic regression are:

1. **Dependent variable:** The dependent variable in logistic regression is binary or categorical, typically coded as 0 or 1, where 0 represents the absence of an event and 1 represents the presence of an event.

2. **Probability modeling:** Logistic regression models the probability of the dependent variable being 1 (the event occurring) as a function of the predictor variables. The model estimates the probability of the event occurring, which ranges between 0 and 1.

3. **Logit transformation:** Logistic regression uses the logit transformation to model the relationship between the predictor variables and the probability of the event occurring. The logit is the natural logarithm of the odds ratio, which is the ratio of the probability of the event occurring to the probability of the event not occurring.

Logistic regression is used in a variety of applications, including:

- Predicting the likelihood of a customer making a purchase (e.g., in e-commerce)
- Determining the risk of a patient developing a certain disease (e.g., in medical diagnosis)
- Classifying email messages as spam or not spam (e.g., in email filtering)
- Estimating the probability of a loan applicant defaulting on a loan (e.g., in credit risk assessment)
- Predicting the outcome of a political election (e.g., in political science research)

The key advantage of logistic regression is its ability to handle binary or categorical dependent variables, which are common in many real-world applications. It provides a straightforward way to estimate the probability of an event occurring and to understand the relationship between the predictor variables and the outcome.

7.a) Describe various Accuracy Measures for Classification.

Ans.7.a) Various accuracy measures for classification include:

1. **Misclassification Error**:
   - Error occurs when a record is classified as belonging to one class when it actually belongs to another.
   - Error Rate = Percentage of misclassified records out of the total records in the validation data.

2. **Confusion Matrix**:
   - A matrix that summarizes the performance of a classification model by showing the counts of true positives, true negatives, false positives, and false negatives.
   - Helps in calculating various accuracy measures.

3. **Error Rate**:
   - Overall error rate is calculated as the sum of false positives and false negatives divided by the total number of instances.
   - Accuracy = 1 - Error Rate.

4. **Sensitivity (True Positive Rate)**:
   - Measures the proportion of actual positive instances correctly identified as positive by the model.
   - Sensitivity = True Positives / (True Positives + False Negatives).

5. **Specificity (True Negative Rate)**:
   - Measures the proportion of actual negative instances correctly identified as negative by the model.
   - Specificity = True Negatives / (True Negatives + False Positives).

6. **Precision**:
   - Precision measures the accuracy of positive predictions.
   - Precision = True Positives / (True Positives + False Positives).

7. **Recall (Sensitivity)**:
   - Recall indicates the proportion of actual positives that were correctly identified.
   - Recall = True Positives / (True Positives + False Negatives).

8. **F1-Score**:
   - The F1-Score is the harmonic mean of precision and recall, providing a balance between the two metrics.
   - F1-Score = 2 * (Precision * Recall) / (Precision + Recall).

These accuracy measures are essential in evaluating the performance of classification models, providing insights into how well the model is classifying instances and helping in selecting the best model based on the specific requirements of the problem.

7.b) Explain in detail the effect -Interaction of the independent variable to the regression model.

Ans.7.b) To explain the effect of the interaction of independent variables on a regression model, we need to consider how the relationship between the independent variables and the dependent variable changes when they interact with each other.

In regression analysis, the interaction effect refers to the situation where the effect of one independent variable on the dependent variable depends on the value of another independent variable.

When there is an interaction between independent variables in a regression model:

- The impact of one independent variable on the dependent variable is not constant but varies based on the value of another independent variable.
- The relationship between the independent variables and the dependent variable is not additive but multiplicative or nonlinear.

This interaction effect can significantly influence the interpretation of the regression model:

- It allows for a more nuanced understanding of how different variables collectively affect the outcome.
- It can reveal hidden relationships that would not be captured by considering each independent variable in isolation.
- It helps in identifying situations where the combined effect of variables differs from what would be expected based on their individual effects.

In summary, the interaction of independent variables in a regression model introduces complexity by showing that the relationship between variables is not always straightforward and can be influenced by the interplay between different factors. This highlights the importance of considering interactions to gain a more comprehensive understanding of the relationships within the data.

8.a) Describe the Chi-square test of independence.

Ans.8.a) The Chi-square test of independence is a statistical test used to analyze the relationship between two categorical variables. It is used to determine whether the two variables are independent or related.

The key steps in the Chi-square test of independence are:

1. Set up the null and alternative hypotheses:
   - H0: The two variables are independent
   - H1: The two variables are not independent

2. Construct a contingency table that shows the observed frequencies for each combination of the two variables.

3. Calculate the expected frequencies under the null hypothesis that the variables are independent.

4. Calculate the Chi-square test statistic using the formula:χ^2 = Σ (O - E)^2 / EWhere:
   - O = Observed frequency
   - E = Expected frequency

5. Determine the degrees of freedom, which is calculated as (number of rows - 1) × (number of columns - 1).

6. Compare the calculated Chi-square statistic to the critical value from the Chi-square distribution table at the desired significance level and degrees of freedom.

7. If the calculated Chi-square statistic is greater than the critical value, reject the null hypothesis and conclude that the two variables are not independent. Otherwise, do not reject the null hypothesis and conclude that the two variables are independent.

The Chi-square test of independence is useful in situations where you want to analyze the relationship between two categorical variables and determine whether they are independent or related. It is a widely used statistical test in various fields, such as social sciences, marketing, and healthcare.

8.b) Give the metrics for computation of distances between the objects.

Ans.8.b) The metrics commonly used for the computation of distances between objects in cluster analysis are:

1. **Euclidean Distance:**
   - The Euclidean distance is the most common distance metric used in cluster analysis.
   - It calculates the straight-line distance between two points in a multidimensional space.
   - The formula for Euclidean distance between two points
   - $P=(p1,p2,...,pn)P=(p1,p2,...,pn)$ and $Q=(q1,q2,...,qn)Q=(q1,q2,...,qn)$ is:
   - $\boldsymbol{Euclidean\ Distance} = \sqrt{(\boldsymbol{p1}-\boldsymbol{q1})^2 + (\boldsymbol{p2}-\boldsymbol{q2}) + \ldots + (\boldsymbol{pn}-\boldsymbol{qn})^2}$

2. **Manhattan Distance (City Block Distance):**
   - The Manhattan distance is the sum of the absolute differences between the coordinates of two points.
   - It is calculated as the sum of the absolute differences in the coordinates along each dimension.
   - The formula for Manhattan distance between two points PP and QQ is:
   - $Manhattan\ Distance = \mid p_1 - q_1 \mid + \mid p_2 - q_2 \mid + \ldots + \mid p_n - q_n \mid$

3. **Minkowski Distance:**
   - The Minkowski distance is a generalization of both the Euclidean and Manhattan distances.
   - It is defined as:
   - $Minkowski\ Distance = (\sum_{i=1}^{n} \mid p_i - q_i \mid^r)1/r$
   - When $r=1r=1$, it reduces to the Manhattan distance, and when $r=2r=2$, it becomes the Euclidean distance.

4. **Cosine Similarity:**
   - Cosine similarity measures the cosine of the angle between two vectors in a multidimensional space.
   - It is often used in text mining and recommendation systems.
   - The formula for cosine similarity between two vectors AA and BB is:
   - Cosine Similarity=$A \cdot B \| A \| \| B \|$
   - Cosine Similarity=$\| A \| \| B \| A \cdot B$

These distance metrics play a crucial role in determining the similarity or dissimilarity between objects in cluster analysis, helping to group similar objects together and separate dissimilar ones.

9.a) What do you mean by categorical variable ? how to find dissimilarity between categorical variables?

Ans.9.a) Categorical variables are a type of variable that can take on more than two states or categories. Unlike binary variables which have only two possible states, categorical variables can have multiple distinct states or categories.

Some key points about categorical variables:

- A categorical variable is a generalization of a binary variable, where the variable can take on more than two states.
- The states of a categorical variable can be denoted by letters, symbols, or a set of integers, but these integers do not represent any specific ordering.
- To compute the dissimilarity between two objects described by categorical variables, the following approach can be used:

1. Let the number of states (categories) of the categorical variable be M.
2. The dissimilarity between two objects i and j can be computed based on the ratio of mismatches:
   $d(i, j) = (p - m) / p$
   Where:
   - m is the number of matches (i.e., the number of variables for which i and j are in the same state)
   - p is the total number of variables
3. Weights can be assigned to increase the effect of m or to assign greater weight to the matches in variables having a larger number of states.

For example, if we have a categorical variable "test-1" with 4 possible states, and we want to find the dissimilarity between two objects 2 and 1, where object 2 is in state 3 and object 1 is in state 1, the dissimilarity would be calculated as:
$d(2,1) = (4 - 0) / 4 = 1$

This shows that the dissimilarity between the two objects is 1, since they are in different states for the single categorical variable.

9.b) Compare agglomerative and divisive hierarchical clustering methods.

Ans.9.b)

Criteria	Agglomerative Hierarchical Clustering	Divisive Hierarchical Clustering
Approach	Bottom-up strategy that starts with each object in its own cluster and successively merges the closest clusters until all objects are in one cluster.	Top-down strategy that starts with all objects in one cluster and recursively splits clusters until each object is in its own cluster.
Process	Begins with n clusters (where n is the number of objects) and iteratively merges the two closest clusters, reducing the number of clusters by one in each step.	Begins with one cluster containing all objects and iteratively splits the cluster with the highest dissimilarity until each object is in its own cluster.
Advantages	- Computationally less complex - Can handle non-convex and irregularly shaped clusters	- Can potentially correct mistakes made in earlier splitting decisions - More flexible in handling different cluster shapes
Disadvantages	Once a merge decision is made, it cannot be undone, leading to potential errors in the final clustering.	- Computationally more complex, especially for large datasets, as it requires evaluating all possible splits at each step.
Suitability	Suitable when the number of clusters is not known in advance and the focus is on exploring the hierarchical structure of the data.	Suitable when the number of clusters is not known in advance and the focus is on identifying the optimal number of clusters through a top-down approach.

The choice between the agglomerative and divisive hierarchical clustering methods depends on the specific characteristics of the data and the goals of the clustering analysis. Agglomerative methods are generally more efficient and easier to implement, while divisive methods offer more flexibility in handling complex cluster shapes but are computationally more demanding.

10.a) What are different attribute selection measures in decision tree?

Ans.10.a) Decision tree attribute selection measures are used to determine the best attribute to split the data at each node in the tree. The three most popular attribute selection measures are information gain, gain ratio, and Gini index.

1. **Information gain**

- Information gain is a measure of the difference in entropy before and after the split.
- It is calculated as the difference between the entropy of the parent node and the weighted average of the entropy of the child nodes.
- The attribute with the highest information gain is selected as the splitting attribute.

2. **Gain Ratio**

- Gain ratio is a modification of information gain that takes into account the number of categories of the attribute.
- It is calculated as the ratio of information gain to the intrinsic information of the attribute.
- The attribute with the highest gain ratio is selected as the splitting attribute.

3. **Gini index**

- Gini index is a measure of the impurity of the data. It is calculated as the sum of the squared probabilities of each category.
- The attribute with the lowest Gini index is selected as the splitting attribute.

These measures are used to build decision trees that are accurate and easy to interpret. The choice of measure depends on the nature of the data and the problem at hand.

10.b) What is CART?

Ans.10.b) CART (Classification and Regression Trees) is a decision tree algorithm used for both classification and regression problems. It is a popular machine learning algorithm used for building decision trees based on the values of features in a dataset. The algorithm splits the data based on the feature that provides the best split, according to a certain criterion such as information gain or Gini impurity.

CART creates binary trees, where each internal node represents a feature, each branch represents a decision rule, and each leaf node represents a class label or a continuous value. The algorithm uses a greedy approach to find the best split at each node, without considering the future splits.

CART can handle both categorical and numerical data, making it a versatile algorithm for various types of data analysis tasks. It can also handle missing values by using surrogate splits, which are alternative splits that can be used when the primary split is not applicable due to missing values.

CART has several advantages, such as its simplicity, interpretability, and ability to handle both types of data. However, it can also suffer from overfitting, where the tree becomes too complex and captures the noise in the data. To prevent overfitting, CART uses techniques such as pruning, where the tree is reduced to a smaller size by removing the less important nodes.

In summary, CART is a powerful decision tree algorithm that can be used for both classification and regression problems. It is simple, interpretable, and versatile, but it requires careful tuning to prevent overfitting and ensure good performance.

10.c) How does tree pruning work in decision tree?

Ans.10.c) Tree pruning is a technique used in decision trees to reduce the size of the tree and prevent overfitting. It involves removing some of the branches of the tree that are based on noise or outliers in the training data.

There are two common approaches to tree pruning: pre-pruning and post-pruning.

In pre-pruning, the tree is pruned by halting its construction early. This can be done by assessing the goodness of a split using statistical measures such as statistical significance, information gain, or Gini index. If the split is not good enough, the algorithm stops splitting at that node.

In post-pruning, the tree is fully grown first, and then subtrees are removed from it. A subtree at a given node is pruned by replacing it with a leaf node that is labeled with the most frequent class among the subtree being replaced.

Pruned trees tend to be smaller and less complex than unpruned trees, making them easier to comprehend and faster at correctly classifying independent test data. However, they may have higher error rates than shallow trees with many leaves.

Several comparative studies have found that no one attribute selection measure has been found to be significantly superior to others, and that measures that tend to produce shallower trees may be preferred.